

Windowed Minority Guidance: Timestep-Localized Effects in Diffusion Denoising Across a Six-Scale Guidance Sweep

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Abstract

Minority guidance [6] steers diffusion sampling toward under-represented data regions via a classifier gradient at every denoising step. We test whether this effect is timestep-localized, comparing three equal-thirds windows named by timestep index (not DDPM chronological order): early $t \in [0, 333)$ (low noise, fine-detail refinement), mid $t \in [333, 667)$, late $t \in [667, 1000)$ (high noise, coarse-structure formation)—against full-chain and no-guidance baselines over 50 seeds on LSUN Bedroom, repeated across six guidance scales $s \in \{0.5, 1.0, 2.0, 3.5, 5.0, 7.0\}$ (including Um et al.’s operating scale $s = 3.5$; 1500 runs in the sweep, plus a sliding-window and a replication study). The mid window has the largest single-window relative effect at every scale (increasing across the sweep from $\Delta_{\text{rel}} = 0.456$ at $s = 0.5$ to 0.832 at $s = 7.0$, after a near-flat $s \leq 1.0$ segment), and the early window achieves lower classifier loss than baseline on all 50 seeds at every scale ($W = 0$, $r_{\text{rb}} = 1.000$; one sample per seed, with modest per-seed magnitude—pooled Cohen’s $d = 0.14$ at $s = 0.5$, growing with scale). The late-window effect is not significant under the per-scale Bonferroni threshold $\alpha/7 = 0.00714$ for $s \leq 1.0$ but is significant from $s = 2.0$ upward; mid’s advantage over the late window is likewise significant only for $s \geq 2.0$ (at $s \leq 1.0$ the two are statistically indistinguishable), and the late-vs-early difference is never significant. A 250-step sliding window at $s = 3.5$ peaks at $t \in [375, 625)$ with a unimodal profile. The mid Δ_{rel} matches on independent seeds 51–100 (0.693 vs. 0.693), though both baseline and mid means shifted ($11.978 \rightarrow 12.759$ and $4.31 \rightarrow 4.60$ respectively), so the match reflects similarly-shifted numerator and denominator rather than absolute-effect reproduction. The relative-effect comparison across scales is partly denominator-sensitive (the full-chain mean loss approaches zero as s rises, making the denominator (baseline – full) grow, with ratio instability worst at low s) and is association, not a causal decomposition; results are a single dataset, architecture, one sample per seed, and classifier loss is a proxy rather than a perceptual metric.

1 Introduction

Classifier-guided diffusion models [1] steer generation with classifier gradients at every denoising step. Um et al. [6] extend this to minority guidance, directing sampling toward underrepresented data regions. Because decreasing timesteps transition from coarse-structure formation to fine-detail refinement [3], the mid-chain phase may be most receptive to minority gradient signals—enabling cheaper inference-time guidance if guidance there recovers most of the full-chain effect. Kynkäänniemi et al. [4] recently showed that classifier-free guidance (CFG) applied only in a middle interval improves FID on class-conditional ImageNet, with guidance harmful at high noise and unnecessary at low noise. We investigate whether an analogous localization holds for the minority classifier guidance of Um et al., which operates via an external minority-score signal rather than a conditional score gap. We test this directly: the original single-scale experiment [5] (Section 3) reported findings that partially challenge it, and its stated limitation was that results might not generalize to other scales. This report addresses that limitation directly, repeating the five-condition experiment across six guidance scales (Section 4), refining the window granularity with a sliding window (Section 5), and probing sample-size and seed robustness (Section 6).

2 Experiment

We use the LSUN Bedroom 256×256 diffusion model with the frozen minority classifier of Um et al. [6] (mc_lsun.pt) stacked on top of OpenAI’s ImageNet feature extractor (256x256_classifier.pt), timestep_respacing=250.

Class 99: Um et al. assign images to 100 ordinal minority-score buckets; class 99 is the rarest (most minority) bucket, not a semantic label. Five conditions: baseline, minority-full $t \in [0, 1000)$, minority-early $t \in [0, 333)$, minority-mid $t \in [333, 667)$, minority-late $t \in [667, 1000)$. Guidance is rescaled to the active window only; results may not generalize. The original experiment ran 50 seeds $\times$ 5 conditions = 250 runs at guidance_scale=1.0 (Kaggle P100, float32). Seeds 1–10 used for protocol verification (no systematic bias observed); the remaining seeds follow the same configuration, which is based on the protocol, checkpoints, and class-id of Um et al. [6].

Sweep extension. To examine scale sensitivity, the same five-condition protocol is repeated at five additional scales $s \in \{0.5, 2.0, 3.5, 5.0, 7.0\}$, each 50 seeds $\times$ 5 conditions, and combined with the original $s = 1.0$ run for a six-scale sweep ($s \in \{0.5, 1.0, 2.0, 3.5, 5.0, 7.0\}$). The five new scales were chosen to bracket Um et al.’s released LSUN Bedroom sampling configuration ($s = 3.5$) on both sides; this scale set was fixed before the sweep was run and was not adapted to the results. Shared baseline: the same 50 seeds at guidance_scale=0 serve as baseline for all six scales; cross-scale comparisons are between within-scale condition pairs, not across baselines.

Primary metric: classifier cross-entropy loss at class 99 (lower = stronger guidance). Softmax confidence is corroborated only—at $s = 1.0$ the single-scale confidence distributions are severely right-skewed (mean/median: full 108x; late 154x; mid 46x; early 58x). Switching from the originally planned metric (classifier confidence) to loss is an unplanned procedural deviation; classifier-based effect sizes are one order of magnitude smaller.

Timestep convention. Window names follow the numerical magnitude of the timestep index t , not the chronological order of the DDPM reverse process [3]. “Early” ($t \in [0, 333)$) therefore corresponds to low-noise timesteps (fine-detail refinement); “late” ($t \in [667, 1000)$) to high-noise timesteps (coarse-structure formation).

Windowed cond_fn. Guidance is applied exclusively within the designated timestep window. Outside the window, the cond_fn is a no-op, returning a zero tensor. Full details are given in Appendix A.2.

3 Results: Single Scale ($s = 1.0$)

Table 1 ($n = 50$; lower loss = stronger guidance).

Table 1: Condition statistics ($n = 50$ seeds; scale = 1.0; LSUN Bedroom). Lower mean loss indicates stronger minority-class guidance. Win Rate is the fraction of seeds on which the condition beats baseline. Relative Effect = $(\text{baseline} - \overline{\text{cond}}) / (\text{baseline} - \overline{\text{full}})$; not a causal decomposition (windows interact across the rejection trajectory).

Condition	Mean Loss	CV	Win Rate	Relative Effect
baseline	11.978	0.19	—	0.000
minority-early	11.129	0.23	50/50	0.162
minority-mid	9.591	0.35	39/50	0.456
minority-late	10.717	0.30	34/50	0.241
minority-full	6.741	0.61	44/50	1.000

The rank ordering mid > late > early > baseline holds across 50 seeds ($r_{\text{rb}} = 1.000, 0.812, 0.432, 0.925$ for early, mid, late, full vs. baseline respectively). Table 2 reports Wilcoxon signed-rank tests, Bonferroni threshold $\alpha/7 = 0.00714$.

Mid recovers 0.456 of full-chain loss reduction (mean 9.591 vs. baseline 11.978 and full 6.741). Analysis criterion: Bonferroni-corrected $p < \alpha/7 = 0.00714$. Bootstrap 95% CI on mid mean: [8.67, 10.50]. The relative effect ($\Delta_{\text{rel}} = 0.456$) measures fraction of full-chain loss reduction recovered; the CI bounds are absolute loss units and are not directly comparable to Δ_{rel} .

Table 2: Wilcoxon signed-rank tests ($n = 50$ pairs; two-sided; zero_method=zsplitt). $r_{rb} \in [0, 1]$: rank-biserial correlation (effect size); $r_{rb} > 0$ indicates first condition has lower loss. Survives Bonferroni correction if $p < \alpha/7 = 0.00714$. The family of seven comprises the four window-vs-baseline comparisons plus the three cross-window pairs (mid-vs-late, late-vs-early, full-vs-mid) that directly bear on the localization claim; the three remaining pairs (full-vs-early, full-vs-late, mid-vs-early) compare conditions with well-established ordering and were excluded from the pre-specified family.

Comparison	W	p -value	r_{rb}
mid vs. baseline	120	< 0.0001	0.812
late vs. baseline	362	0.00716	0.432
early vs. baseline	0	< 0.0001	1.000
full vs. baseline	48	< 0.0001	0.925
mid vs. late	391	0.0166	0.387
late vs. early	547	0.3882	0.142
full vs. mid	181	< 0.0001	0.716

Key finding: at $s = 1.0$, early-window guidance (operating at low noise, fine-detail phase) reduces classifier loss on all 50 seeds ($W = 0$, $r_{rb} = 1.000$, $p < 0.0001$); confidence corroborates this direction only. We measured no perceptual or quality metric, so this is a statement about the classifier-loss objective, not about minority-class image quality, and we draw no causal conclusion.

Late vs. Early at scale= 1.0: the coarse-structure (late) window has a higher Δ_{rel} point estimate than the fine-detail (early) window at this scale, though the difference is not statistically significant ($p = 0.388$).

4 Results: Multi-Scale Sweep

We repeat the five-condition experiment across six guidance scales $s \in \{0.5, 1.0, 2.0, 3.5, 5.0, 7.0\}$ (50 seeds each). Table 3 reports the cross-scale relative-effect matrix and Figure 1 plots Δ_{rel} against scale for each window. Three patterns hold across the sweep: the mid window has the largest single-window relative effect at every scale; the early window beats baseline on all 50 seeds at every scale; and the late window becomes significant only from $s = 2.0$ upward.

Table 3: Cross-scale relative-effect matrix ($\Delta_{rel} = (\text{baseline} - \overline{\text{cond}})/(\text{baseline} - \overline{\text{full}})$; $n = 50$ seeds per scale; LSUN Bedroom). The full condition is 1.000 by construction at every scale. Last column: 95% bootstrap CI on the mid mean loss (absolute units; independent-sample resampling of mid losses, 10,000 resamples, random.seed(42); not directly comparable to the Δ_{rel} CIs in Figure 1, which use paired-seed resampling of Δ_{rel} via np.random.default_rng(42)).

Scale s	early	mid	late	mid mean-loss CI
0.5	0.176	0.456	0.149	[10.37, 11.81]
1.0	0.162	0.456	0.241	[8.67, 10.50]
2.0	0.242	0.544	0.294	[5.82, 8.02]
3.5	0.379	0.693	0.405	[3.28, 5.41]
5.0	0.494	0.774	0.500	[2.13, 4.04]
7.0	0.608	0.832	0.566	[1.40, 3.24]

Mid-window relative effect is largest at every scale. The mid window’s relative effect rises with scale (effectively flat between $s = 0.5$ and $s = 1.0$, then increasing)—0.456 ($s = 0.5$), 0.456 ($s = 1.0$), 0.544 ($s = 2.0$), 0.693 ($s = 3.5$), 0.774 ($s = 5.0$), 0.832 ($s = 7.0$)—and has the largest point estimate among the three windows

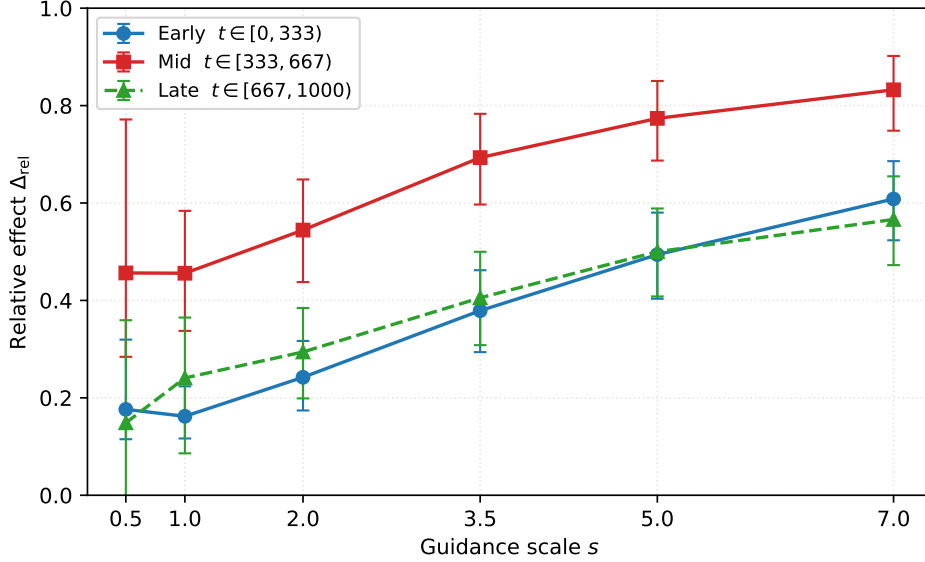


Figure 1: Relative effect Δ_{rel} versus guidance scale s for the early, mid, and late windows. Mid has the largest relative effect at every scale; early and late track each other closely and are never significantly different (late-vs-early $p \geq 0.357$ at every scale, minimum at $s = 2.0$), with overlapping CIs throughout, so the apparent changes in their relative ordering across scales are not statistically meaningful. The x -axis is linear in s ; error bars are 95% bootstrap CIs on Δ_{rel} (paired-seed resampling, 10,000 resamples). The late-window CI at $s = 0.5$ spans zero, consistent with its non-significance there.

at every scale. Mid vs. baseline survives the per-scale Bonferroni threshold ($\alpha/7 = 0.00714$) at all six scales (e.g. $W = 120$, $p < 0.0001$ at $s = 1.0$; $W = 9$, $p < 0.0001$ at $s = 3.5$; $W = 0$, $p < 0.0001$ at $s = 5.0$ and $s = 7.0$). Mid’s lead over the late window, however, is statistically significant only for $s \geq 2.0$ (mid-vs-late $p = 9.9 \times 10^{-5}$ at $s = 2.0$, down to 1.4×10^{-5} at $s = 7.0$); at $s = 0.5$ ($p = 0.082$) and $s = 1.0$ ($p = 0.0166$, failing $\alpha/7$) mid has the larger point estimate but its superiority over the late window is not statistically established. So mid-window “dominance” is a tested claim for $s \geq 2.0$ and a point-estimate ordering at the two lowest scales. The rising Δ_{rel} should be read with care: because Δ_{rel} uses (baseline – full) as its denominator and the full-chain mean loss approaches 0 as s grows (full mean 10.05 at $s = 0.5$ down to 0.281 at $s = 7.0$), the cross-scale comparison is partly denominator-sensitive and is an association, not a causal decomposition of where guidance acts.

Late-window significance transition. The late window does not survive the per-scale Bonferroni threshold for $s \leq 1.0$ (late vs. baseline $p = 0.163$ at $s = 0.5$; $p = 0.00716$ at $s = 1.0$, narrowly above $\alpha/7 = 0.00714$) but is significant from $s = 2.0$ upward ($p = 4.03 \times 10^{-6}$ at $s = 2.0$; $p = 7.24 \times 10^{-10}$ at $s = 3.5$; $p < 10^{-11}$ at $s = 5.0$ and $s = 7.0$). The late-window relative effect rises correspondingly from 0.149 to 0.566 across the sweep.

Early window: all seeds improve at every scale. The early window beats baseline on all 50 seeds (one sample per seed) at every scale: early vs. baseline gives $W = 0$, $r_{\text{rb}} = 1.000$, win rate 50/50, $p < 0.0001$ at each of the six scales. This is an extreme but consistently observed result (all 50 paired differences favor the early window); we report it as the observed win rate rather than interpreting it as a causal effect. Per-seed reductions are reliable in direction but modest when measured as pooled Cohen’s d ($d = 0.14$ at $s = 0.5$, growing with scale); the paired d_z (mean paired difference divided by its SD) is larger ($d_z = 1.02$ at $s = 0.5$) because it removes between-seed variance shared across conditions.

Late vs. early is never significant. Across all six scales the late-vs-early comparison never survives Bonferroni correction: $p = 0.759$ ($s = 0.5$), 0.388 ($s = 1.0$), 0.357 ($s = 2.0$), 0.585 ($s = 3.5$), 0.992 ($s = 5.0$), 0.432 ($s = 7.0$). The two windows track each other closely (Figure 1): their Δ_{rel} point estimates change relative order across scales (early higher at $s = 0.5$, late higher through the mid scales, early higher again at $s = 7.0$) and are nearly equal at $s = 5.0$ (Δ_{rel} : early 0.494 vs. late 0.500), but with overlapping bootstrap CIs and no significant test at any scale, these order changes are not evidence that one window outperforms the other.

Cross-scale multiple comparisons. The seven within-scale comparisons are corrected at $\alpha/7 = 0.00714$ as in the single-scale analysis; this per-scale correction is unchanged by adding scales. Pooling the six scales raises a multiple-comparisons question across the full $6 \times 7 = 42$ comparisons; under the stricter cross-scale threshold $\alpha/42 = 0.00119$ the late-vs-baseline result at $s = 1.0$ ($p = 0.00716$) and at $s = 0.5$ ($p = 0.163$) remain non-significant, while the mid-, early-, and full-vs-baseline results and the late-vs-baseline results for $s \geq 2.0$ continue to pass. This $\alpha/42$ family covers the main sweep’s 6×7 condition comparisons; the sliding-window is a separate localization follow-up (corrected within itself at $\alpha/8$) and the sample-size sweep is a post-hoc robustness probe (corrected at $\alpha/7$ within each n), rather than folded into the sweep family.

5 Results: Sliding Window

To refine the three coarse equal-thirds windows, we sweep a 250-step window of constant width across the chain at $s = 3.5$ —the operating scale of Um et al. [6]—stepped by 125 raw timesteps (slide_0: $t \in [0, 250)$ through slide_6: $t \in [750, 1000)$; seven windows, 50 seeds each). All seven sliding windows beat baseline under a Bonferroni threshold computed for this experiment’s eight comparisons (seven windows plus full), $\alpha/8 = 0.00625$ —separate from the main sweep’s $\alpha/7$. The relative-effect profile is unimodal (Figure 2), peaking at slide_3 ($t \in [375, 625)$) with $\Delta_{\text{rel}} = 0.536$ (mean loss 6.050, $W = 8$, $p < 0.0001$, $r_{\text{rb}} = 0.987$; Table 4).

Table 4: Sliding-window results at $s = 3.5$ (250-step window, step 125; $n = 50$; LSUN Bedroom). $\Delta_{\text{rel}} = (\text{baseline} - \text{cond})/(\text{baseline} - \text{full})$ with baseline mean 11.978 and full-chain mean 0.912 (denominator = 11.066). Bonferroni threshold $\alpha/8 = 0.00625$ (eight comparisons). All windows are significant. Error bars are 95% bootstrap CIs on Δ_{rel} (paired-seed resampling, 10,000 resamples); triangles mark the equal-thirds 334-step windows (early/mid/late) at $s = 3.5$ for comparison—wider than the 250-step sliding windows, hence the higher mid value.

Window	$[t_{\text{start}}, t_{\text{end}})$	Mean Loss	Δ_{rel}	r_{rb}
slide_0	$[0, 250)$	9.711	0.205	1.000
slide_1	$[125, 375)$	6.994	0.450	1.000
slide_2	$[250, 500)$	6.833	0.465	0.984
slide_3	$[375, 625)$	6.050	0.536	0.987
slide_4	$[500, 750)$	6.567	0.489	0.842
slide_5	$[625, 875)$	7.322	0.421	0.856
slide_6	$[750, 1000)$	8.754	0.291	0.758

The peak window is centred at $t = 500$ and lies fully inside the equal-thirds mid range $[333, 667)$ (the peak ends at $t = 625$, 42 steps before the mid/late boundary), consistent with the mid-window dominance seen in the sweep. The sliding windows are 250 raw timesteps wide, whereas the equal-thirds mid window is 334 raw timesteps wide; accordingly the sliding peak slide_3 ($t \in [375, 625)$) reaches $\Delta_{\text{rel}} = 0.536$, lower than the 334-step equal-thirds mid window’s $\Delta_{\text{rel}} = 0.693$ at the same scale $s = 3.5$. This gap is consistent with the narrower sliding width covering less of the receptive interval, not with a contradiction or a different effective region. As with Δ_{rel} elsewhere, this is an association across overlapping windows, not a causal decomposition.

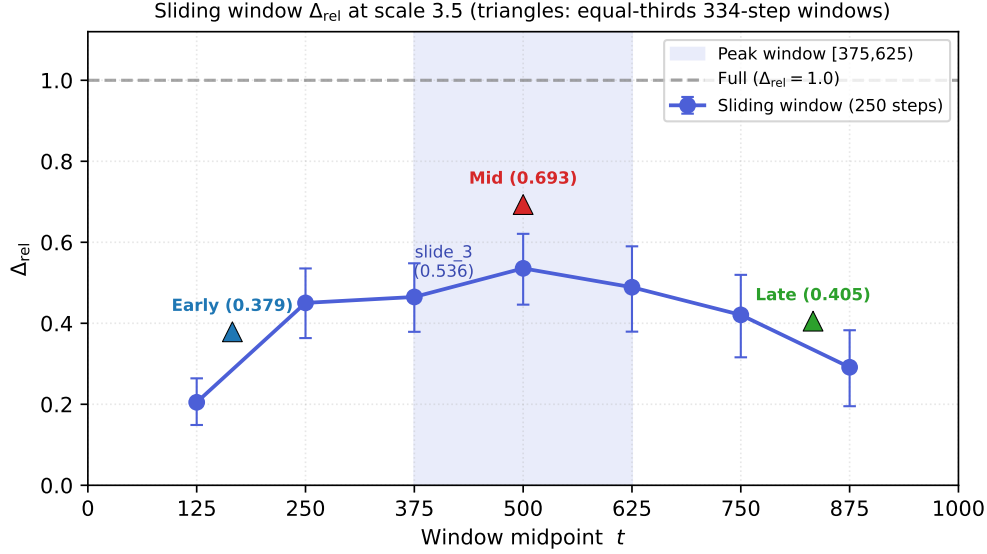


Figure 2: Sliding-window relative effect at $s = 3.5$ (window width 250, step 125) plotted against the window midpoint t . The profile is unimodal with a peak at slide_3 ($t \in [375, 625]$, midpoint 500, $\Delta_{\text{rel}} = 0.536$). Triangles mark the original equal-thirds early/mid/late reference windows.

6 Results: Robustness

Sample-size sensitivity. The single-scale concern is the late-window result at $s = 1.0$, which sits at the Bonferroni boundary. Recomputing the late-vs-baseline Wilcoxon test on the first n seeds in seed order: $p = 0.113$ at $n = 25$ (fails), $p = 0.00665$ at $n = 35$ (passes $\alpha/7 = 0.00714$), and $p = 0.00716$ at $n = 50$ (fails). The result is fragile: it does not consolidate as n increases from 35 to 50, which is atypical of a true effect (which would be expected to strengthen or hold as n grows). We treat the $s = 1.0$ late result as genuinely marginal and do not claim significance. The mid and early windows pass at every $n \in \{25, 35, 50\}$ for every scale, and from $s = 2.0$ upward the late window passes at every n .

Independent replication (seeds 51–100). We re-ran the five conditions at $s = 3.5$ on independent seeds 51–100 (50 new seeds). The mid-window relative effect replicates closely: $\Delta_{\text{rel}} = 0.693$ on seeds 51–100, matching the 0.693 obtained on seeds 1–50. The mid window again beats baseline on all 50 seeds ($W = 0$, $r_{\text{tb}} = 1.000$, $p < 0.0001$), as does the early window ($W = 0$, $r_{\text{tb}} = 1.000$), and the late window remains significant ($W = 65$, $p = 3.13 \times 10^{-10}$, $r_{\text{tb}} = 0.898$). It is the mid finding specifically that replicates: the mid relative effect is unchanged ($0.693 \rightarrow 0.693$), whereas the early and late relative effects drift across the independent seed set (early $0.379 \rightarrow 0.334$, late $0.405 \rightarrow 0.492$) and the underlying means shift (baseline $11.978 \rightarrow 12.759$, mid $4.31 \rightarrow 4.60$). The mid relative-effect match (0.693) thus reflects similarly shifted numerator and denominator, not a literal absolute-effect reproduction; the metric is also seed-set sensitive for the weaker (early, late) windows. We do not claim a full replication of the per-window effect sizes, only that the mid-window relative effect reproduces tightly.

7 Discussion and Conclusion

Across the six-scale sweep the mid window has the largest single-window relative effect at every scale; its lead over the late window is statistically significant for $s \geq 2.0$ (not at $s \leq 1.0$), and its relative effect rises with guidance scale (Table 3, Figure 1). All three windows produce detectable seed-wise effects at the operating scales, contradicting strict binary localization: the early window beats baseline on all 50 seeds at every scale ($r_{\text{tb}} = 1.000$, 50/50 wins), and the late window—not significant for $s \leq 1.0$ —becomes significant

from $s = 2.0$ upward. The sliding-window analysis at $s = 3.5$ refines this picture: the relative-effect profile is unimodal and peaks at the center of the chain ($t \in [375, 625)$, $\Delta_{\text{rel}} = 0.536$), consistent with mid-chain dominance without a sharp boundary.

The cross-scale pattern—mid above early and late in point estimate at every scale (the mid-vs-late gap significant for $s \geq 2.0$), with early and late statistically indistinguishable at every scale (late-vs-early never significant, overlapping CIs)—is anchored on the relative-effect numbers, not on intuition about diffusion dynamics. We emphasize that Δ_{rel} uses (baseline – full) as denominator; the full-chain mean loss approaches 0 as scale rises, so the denominator grows (1.93 at $s = 0.5$ to 11.70 at $s = 7.0$). This means ratio-estimator instability is worst at low s (where the denominator is small and the late-window CI spans zero at $s = 0.5$), not at high s as might be assumed. The cross-scale comparison is still partly denominator-sensitive and is an association, not a causal decomposition.

Limitations: single dataset (LSUN Bedroom), single architecture and minority classifier, class 99, $n = 1$ sample per seed; high variance that grows with scale (full-chain CV 0.61 at $s = 1.0$ rising to 1.04 at $s = 7.0$, the mid window reaching CV 1.49 at $s = 7.0$); no perceptual metrics (FID [2], precision/recall). The sweep adds two limitations. First, pooling six scales raises a cross-scale multiple-comparisons question: we correct within scale at $\alpha/7 = 0.00714$ and note the stricter cross-scale threshold $\alpha/42 = 0.00119$, under which the $s \leq 1.0$ late-vs-baseline results remain non-significant while the mid/early/full results and the $s \geq 2.0$ late results continue to pass. Second, the sweep remains a single dataset; scale generality has been tested but dataset and architecture generality have not. All six scales, the cross-scale matrix, the sliding-window run, and the seeds 51–100 replication reproduce from their released raw CSVs. The sliding-window analysis is a localization follow-up to the sweep’s mid-dominance pattern, and the seed-replication is a confirmatory test of mid-window seed stability; the sample-size sweep is a post-hoc robustness probe. Each is Bonferroni-corrected within itself; we do not fold their tests into a single study-wide family, and a stricter study-wide correction would be more conservative still.

Revised framing: mid has the largest point-estimate relative effect at every scale (statistically established vs. late for $s \geq 2.0$; point-estimate ordering only at $s \leq 1.0$) and the effect is distributed across windows; no single equal-thirds window approximates full-chain performance, and the dominant region is the center of the chain rather than a sharp early/late split. The case for treating the mid window as a cheaper inference-time guidance target is most defensible at higher scales ($s \geq 3.5$), where the mid relative effect reaches 0.693–0.832 and replicates on independent seeds (the match reflects similarly-shifted numerator and denominator, not literal absolute reproduction); even there it recovers a fraction, not the whole, of the full-chain effect.

References

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A Reproducibility

A.1 Experimental Configuration

Models and checkpoints (all publicly available):

Table 5: Model checkpoints used in all experiments.

Component	Checkpoint	Source
Diffusion model	lsun_bedroom.pt	OpenAI guided-diffusion
Feature extractor	256x256_classifier.pt	OpenAI guided-diffusion
Minority classifier	mc_lsun.pt	Um et al. (2024)

Shared hyperparameters (all fixed across conditions; the guidance_scale column lists the six swept values):

Table 6: Hyperparameters shared across all conditions and seeds.

Parameter	Value
diffusion_steps	1000
timestep_respadding	250
noise_schedule	linear
guidance_scale	0.5, 1.0, 2.0, 3.5, 5.0, 7.0 (swept)
image_size	256
num_channels	256
num_res_blocks	2
attention_resolutions	32,16,8
learn_sigma	True
use_fp16	False
manual_class_id	99
num_samples	1

The minority classifier (mc_lsun.pt) expects 8×8 latent features produced by 256x256_classifier.pt with pool=True, in_channels=512, num_channels=100, classifier_width=128, classifier_pool=attention.

UNET.py patch—the feature extractor 256x256_classifier.pt loads as an EncoderUNetModel (not the UNetModel used for diffusion sampling). Its forward is patched to expose the 8×8 pooled feature map:

Listing 1: unet.py patch: expose intermediate pooled features during sampling.

```
1 def forward(self, x, timesteps, f_out=False):
2     # skipping feature map extracted until final return
3     if f_out:
4         return h
5     return self.out(h) # original logic
```

Data schema (one CSV per scale/experiment; one row per run):

run_id		condition		guidance_window		t_start		t_end		guidance_scale		seed		n_generated		classifier_mean_confidence		classifier_mean_loss		runtime_seconds		artifact_path		timestamp
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Metrics are computed at $t=0$ (final denoising step) by passing generated images through the frozen classifier chain. Each run produces one image; classifier_mean_ fields are single-sample values ($n=1$ per seed per condition). The multi-scale sweep adds one 50-seed $\times$ 5-condition CSV per scale; the sliding-window run adds

a 9-condition (baseline, full, slide_0–slide_6) CSV at $s = 3.5$; the replication run adds a 5-condition CSV for seeds 51–100 at $s = 3.5$. The sliding-window and seeds 51–100 replication runs share the configuration above (the appendix model and hyperparameter tables) except for `guidance_scale`, the seed range, and the window boundaries.

A.2 Windowed Guidance Implementation

The windowed guidance requires no architectural modification—it is implemented entirely within the `cond_fn` callback:

Listing 2: Windowed `cond_fn`: applies classifier gradient only within $[t_start, t_end]$.

```

1  t_start = args.t_start    # e.g. 333 for mid window
2  t_end   = args.t_end     # e.g. 667 for mid window
3
4  def cond_func(x, t=None):
5      current_t = t[0].item()    # actual denoising timestep
6      # Zero grad outside the active window
7      if not (t_start <= current_t < t_end):
8          return torch.zeros_like(x)
9      with torch.enable_grad():
10         x_in = x.detach().requires_grad_(True)
11         latents = f_extractor(x_in, timesteps=t, f_out=True) # 8x8 pooled
12         features
13         logits = classifier(latents, timesteps=t)            # minority
14         classifier(mc_lsun.pt)
15         log_probs = F.log_softmax(logits, dim=-1)
16         selected = log_probs[range(log_probs.shape[0]), y.view(-1)]
17         # Scale applied only; sampler must NOT re-apply classifier_scale
18         return torch.autograd.grad(selected.sum(), x_in)[0] \
19             * args.classifier_scale

```

For baseline runs, `guidance_scale=0` is passed rather than gating by timestep. Window boundaries are raw diffusion timesteps (0–1000); the active window for mid is $t \in [333, 667]$, corresponding to 334 of the 1000 raw timesteps (≈ 83 of the 250 respaced steps at stride 4). The sliding-window run uses a constant width of 250 raw timesteps stepped by 125.

A.3 Statistical Analysis Code

Listing 3: Statistical analysis: Wilcoxon signed-rank tests and rank-biserial correlation ($n=50$).

```

1  import csv, random
2  from scipy import stats
3
4  # --- load paired losses ---
5  data = {}
6  with open('data/multi_scale/runs_scale1p0_lsun_bedroom.csv') as f: # schema
7      identical across all scale CSVs
8      for row in csv.DictReader(f):
9          data.setdefault(row['condition'], {})[(int(row['seed']))] = \
10              float(row['classifier_mean_loss'])
11
12  seeds = sorted(data['baseline'])
13  get = lambda cond: [data[cond][s] for s in seeds]
14  baseline, early, mid, late, full = (get(c) for c in
15      ['baseline', 'minority-early', 'minority-mid', 'minority-late', 'minority-full'])
16
17  # --- Wilcoxon signed-rank tests ---
18  def wilcoxon_rbc(a, b):

```

```

18     diffs = [ai - bi for ai, bi in zip(a, b)]
19     # zero_method='zsplitt': split zero ranks equally between + and -
20     W, p = stats.wilcoxon(diffs, zero_method='zsplitt')
21     # rank-biserial: W / (n*(n+1)/2) * 2 - 1 (normalized to [0,1])
22     n = len(diffs)
23     r_rb = 1 - 2 * W / (n * (n + 1) / 2)
24     return W, p, r_rb
25
26     comparisons = [
27         (mid, baseline, 'mid vs. baseline'),
28         (late, baseline, 'late vs. baseline'),
29         (early, baseline, 'early vs. baseline'),
30         (full, baseline, 'full vs. baseline'),
31         (mid, late, 'mid vs. late'),
32         (late, early, 'late vs. early'),
33         (full, mid, 'full vs. mid'),
34     ]
35
36     for a, b, label in comparisons:
37         W, p, r_rb = wilcoxon_rbc(a, b)
38         print(f'{label:<25} W={W:>5.0f} p={p:.4f} r_rb={r_rb:.3f}')
39
40     alpha_corrected = 0.05 / 7 # = 0.00714 (per scale); cross-scale = 0.05/42 = 0.00119
41
42     # --- Bootstrap 95% CI on mid mean (10,000 resamples) ---
43     random.seed(42)
44     boot = sorted([sum(random.choices(mid, k=50)) / 50 for _ in range(10_000)])
45     ci = [boot[249], boot[9749]] # 2.5th and 97.5th percentiles
46
47     baseline_mean = sum(baseline) / len(baseline) # 11.978
48     threshold = 0.05 / 7 # 0.00714
49     mid_mean = sum(mid) / len(mid) # 9.591 at s=1.0

```

Bootstrap note. The code above computes the 95% CI on the mid mean loss (Table 3 last column) via independent resampling of mid losses (random.seed(42), 10,000 resamples). The Δ_{rel} error bars in Figure 1 and Figure 2 use a different method: paired-seed resampling jointly over (baseline, condition, full) losses, implemented via np.random.default_rng(42), with output written to stats/results/delta_rel_cis.json by stats/make_figures.py. These are two distinct quantities (absolute loss vs. relative effect) and use two distinct bootstrap procedures; both use 10,000 resamples and seed 42.

Environment: Python 3.11, scipy 1.x, numpy 2.x. All runs generated on Kaggle (NVIDIA Tesla P100, float32, CUDA 11.x).

Data availability: All raw CSVs, stats scripts, and reproducibility code are in the accompanying repository (<https://github.com/VittoriaLanzo/windowed-minority-guidance-repro>; see data/ subdirectory) and mirrored as a Kaggle dataset (<https://www.kaggle.com/datasets/vittorialanzo/wmg-sweep-results>). The Kaggle notebook that generated the runs is at <https://www.kaggle.com/code/vittorialanzo/windowed-minority-guidance-extended>. All three are currently private and will be made public on acceptance.